

		Change in EPE from X to Y = $EPE_Y - EPE_X = -eV_Y - (-eV_X) = -70 \text{ eV} + 100 \text{ eV} = +30 \text{ eV}$
20	C	$R = \frac{\rho L}{A} \propto \frac{L}{A}$ $\frac{R'}{R} = \frac{L'}{L} \times \frac{A}{A'}$ $\frac{R'}{4.0} = \frac{5d}{d} \times \frac{d^2}{2d(d)}$ $R' = 10 \, \Omega$
21	D	In the driver circuit, the p.d. across XY should be minimized, so the resistance of the